

JSS 3 MATHEMATICS – FIRST TERM

TOPIC 3: ADDITION AND SUBTRACTION OF NUMBERS IN BASE 2

1. Binary Addition Rules

$0 + 0 = 0$; $0 + 1 = 1$; $1 + 0 = 1$; $1 + 1 = 10$ (write 0, carry 1); $1 + 1 + 1 = 11$ (write 1, carry 1). Add column by column from right to left, just as in base 10.

2. Example – Addition

$101 + 111$: $1+1=10$ (write 0, carry 1); $0+1+1=10$ (write 0, carry 1); $1+1+1=11 \rightarrow$ result 1100 .

3. Binary Subtraction

$0 - 0 = 0$; $1 - 0 = 1$; $1 - 1 = 0$; $0 - 1$ requires borrowing. When you borrow, $10 - 1 = 1$. Work from right to left.

4. Practice Tip

Convert to base 10 to check your answer until you are confident. Practise with two or three 3-digit binary numbers as in the scheme.

5. Summary

Binary addition and subtraction follow the same column method as base 10, but with only digits 0 and 1 and carries/borrows in powers of 2.

End of Topic 3